

MOBAN Chapter 7 — Part 3: 5G

Comprehensive Study Guide

PART 1: THEORY SUMMARY

1. What is 5G?

1.1 Definitions

- **Qualcomm:** "5G enables a new kind of network that is designed to connect virtually everyone and everything together including machines, objects, and devices... higher multi-Gbps peak data speeds, ultra-low latency, more reliability, massive network capacity, increased availability, and a more uniform user experience."
- **NGMN Alliance:** "5G is an end-to-end ecosystem to enable a fully mobile and connected society... delivering consistent experience... enabled by sustainable business models."

1.2 Three Service Categories

Category	Abbreviation	Full Name	Target	Key requirement
eMBB	Enhanced Mobile Broadband	High data rates for smartphones, streaming, AR/VR	10–20 Gbps peak; 100 Mbps user	
URLLC	Ultra-Reliable Low-Latency Communications	Industrial automation, autonomous vehicles, remote surgery	< 1 ms latency; 99.999% reliability	
mMTC / MMTC	Massive Machine-Type Communications / Massive IoT	Smart cities, sensors, meters	10 ⁶ devices/km ² ; low power, low cost	

These three categories cannot all be maximised simultaneously — 5G networks use **network slicing** to configure virtual networks optimised for each vertical.

1.3 4G vs. 5G Capability Comparison (from Addestino guest lecture)

Capability	4G (LTE)	5G target
Latency	~100 ms	1 ms
Peak data rate	1 Gbps	10–20 Gbps
Connection density	~10 ⁴ /km ²	10 ⁶ /km ²

Capability	4G (LTE)	5G target
Reliability	~99.99%	99.999%
Speed support	350 km/h	500 km/h

Key insight from Addestino: For warehousing use cases, 4G already meets latency, data rate, and reliability needs. 5G improvements are not yet required for most industrial use cases — 5G's real advantage lies in URLLC + mMTC for future automation and massive IoT.

2. Challenges

The main 5G challenges span multiple axes:

- **Coverage vs. capacity:** Higher frequencies (mmWave) give massive bandwidth but tiny range; lower frequencies give wide coverage but less bandwidth.
- **Heterogeneous networks:** 5G must coexist with 4G, 3G, Wi-Fi, IoT networks.
- **Latency:** Sub-millisecond latency requires processing close to the user (→ MEC).
- **Energy:** Massive MIMO antenna arrays and dense small cells increase power consumption.
- **Spectrum management:** Limited licensed spectrum must be shared across operators and use cases.

3. Mobile Network Services

3.1 eMBB (Enhanced Mobile Broadband)

- Smartphones, hotspots, fixed wireless access
- Video streaming, AR/VR, cloud gaming
- Needs wide channels, massive MIMO, mmWave in dense urban areas

3.2 URLLC (Ultra-Reliable Low-Latency Communications)

- Autonomous vehicles, industrial robotics, remote surgery
- Drone control (ASTRID use case from slides)
- Requires < 1 ms end-to-end latency → short slot structure + MEC
- Requires 99.999% reliability → redundant paths, error correction

3.3 mMTC (Massive Machine-Type Communications)

- Smart city sensors, meters, agriculture monitors
- Low data rate, but massive number of devices
- Battery life measured in years
- Uses NB-IoT or LTE-M as complementary technologies

4. NR (New Radio)

4.1 Spectrum

5G NR uses two frequency ranges:

Range	Name	Frequencies	Characteristics
FR1	Sub-6 GHz	450 MHz – 6 GHz	Good coverage, moderate bandwidth; main 5G bands today
FR2	mmWave	24.25 – 52.6 GHz	Very high bandwidth; short range, high path loss; needs beamforming

FR1 provides coverage; FR2 provides capacity in dense areas.

4.2 Scalable OFDM Numerology

LTE used only 15 kHz subcarrier spacing. 5G NR supports **multiple numerologies** (μ) to serve different use cases:

μ	Subcarrier spacing (kHz)	Slot duration (ms)	Slots per subframe	Use case
0	15	1	1	Compatible with LTE
1	30	0.5	2	Most common FR1
2	60	0.25	4	FR1 and FR2
3	120	0.125	8	FR2 (mmWave)
4	240	0.0625	16	FR2 reference signals only

Fixed durations (like LTE):

- Subframe = **1 ms** (always)
- Radio frame = **10 ms** (always)

Smaller subcarrier spacing → longer symbol duration → better multipath tolerance (rural) Larger subcarrier spacing → shorter slot → lower latency (URLLC, mmWave)

4.3 Frame Structure

```

Radio frame (10 ms)
├── 10 subframes (each 1 ms)
│   ├── N slots per subframe (depends on  $\mu$ )
│   │   ├── 14 OFDM symbols per slot (normal CP)
│   │   └── 12 or 14 OFDM symbols (CP variant)

```

OFDM symbols per slot: **14** (normal cyclic prefix) or **12** (extended CP)

4.4 Slot Formats

5G NR defines **62 slot formats** (+ 194 reserved) at the symbol level (LTE TDD was defined at subframe level).

Symbol types:

- **D** — Downlink only
- **U** — Uplink only
- **X** — Flexible (can be DL or UL)

Transmission rules:

- DL transmission in **D** or **X** symbols
- UL transmission in **U** or **X** symbols

This flexibility is key for TDD: the split between DL and UL can be tuned per slot.

4.5 Massive MIMO and Beamforming

Traditional MIMO (LTE): up to 4×4 antennas. 5G NR uses **massive MIMO** with tens to hundreds of antenna elements.

Two key techniques:

Technique	What it does	Benefit
Beamforming	Focuses transmitted energy in direction of UE	Longer range, less interference, better SNR
Spatial multiplexing	Sends multiple data streams simultaneously to same/different UEs	Higher throughput

Adaptive beamforming for mmWave:

- mmWave has very high path loss → without beamforming, range is too short
- Massive antenna arrays create very narrow beams that track the UE
- Critical for making mmWave practical in mobile networks

5. Architecture

5.1 Cloud RAN (C-RAN) and Open RAN

Traditional cellular RAN: each site has a **BBU** (Baseband Unit) + **RRH** (Remote Radio Head) connected by fronthaul.

C-RAN: Centralize BBU processing in a shared pool. Reduces costs, improves coordination.

Open RAN: Disaggregate the RAN into three logical nodes with open interfaces:

Node	Abbreviation	Location	Handles
Radio Unit	RU	At antenna	RF processing, some L1
Distributed Unit	DU	Near antenna or edge	Real-time L1/L2 processing
Centralized Unit	CU	Central/cloud	Non-real-time L2/L3

Transport links:

- **Fronthaul:** RU ↔ DU (highest bandwidth, strict latency)
- **Midhaul:** DU ↔ CU
- **Backhaul:** CU ↔ Core Network



Abbreviation	Full Name	Role
UE	User Equipment	User terminal
RU	Radio Unit	RF processing at antenna
DU	Distributed Unit	Real-time baseband processing
CU	Centralized Unit	Non-real-time RAN functions
5GC	5G Core	Mobility, routing, policy
NG	Next Generation interface	CU ↔ 5GC interface

5.2 Architecture Trade-offs

Trade-off dimension	Centralized (low split)	Distributed (high split)
Fronthaul bandwidth	High	Low
AP (DU/RU) complexity	Low	High
Processing delay	Non-ideal	Fast (real-time near antenna)
Synchronization accuracy needed	High	Low
Cost	Higher fronthaul	Higher equipment cost

Key insight: Real-time functions (beamforming, HARQ) must stay close to the antenna. Non-real-time functions (scheduling policy, mobility) can be centralized. The "functional split" determines where the DU/CU boundary is.

6. Network Management

6.1 Network Slicing

What: Share physical network resources by creating isolated virtual networks (slices), each individually configured for a specific vertical/use case with guaranteed performance characteristics.

Why needed: eMBB, URLLC, and mMTC have fundamentally different requirements that cannot all be met by a single network configuration.

Example slices:

- Slice A: high bandwidth for video streaming
- Slice B: ultra-low latency for industrial control
- Slice C: low power, massive connections for IoT sensors

6.2 SDN (Software-Defined Networking)

Traditional networking problem: Control Plane (CP) and Data Plane (DP) are tightly coupled inside each device. Each device configured individually via CLI.

SDN solution:

- **Separate CP from DP** — hide device specifics via hardware abstraction
- **Centralize CP** — one logical programmable network view → configuration = function (global view)
- **Northbound API:** application layer talks to controller
- **Southbound API:** controller talks to network devices (standardized, e.g. OpenFlow)

```

Applications
  ↓ Northbound API
SDN Controller (centralized Control Plane)
  ↓ Southbound API
Network devices (Data Plane only – switches, routers)

```

Benefits: Simplified management, programmability, vendor independence, rapid service deployment.

6.3 NFV (Network Function Virtualization)

Traditional: Network functions (firewall, load balancer, NAT) run on dedicated proprietary hardware.

NFV: Run network functions as software on commodity (standard) hardware. Managed by **MANO** (Management and Orchestration).

Benefits: Scale up/down on demand, move functions to optimal location, reduce hardware costs.

6.4 SDN + NFV → Network Slicing

Technology	Contributes
SDN	Centralized control of which resources each slice uses
NFV	Virtualize core network functions; run per-slice instances
Together	Create isolated end-to-end virtual networks over shared physical infrastructure

7. Deployment**7.1 NSA vs. SA**

Mode	Abbreviation	Radio	Core	Description
Non-Standalone	NSA	5G NR + LTE	LTE EPC	Uses existing LTE infrastructure; 5G NR adds capacity. Easier migration.
Standalone	SA	5G NR only	5G Core	Full 5G; required for URLLC and slicing.

- **Dual connectivity:** UE connected to both LTE and 5G NR simultaneously in NSA
- Current deployments (2025): mostly NSA → transitioning to SA

7.2 Mobile Edge Computing (MEC)

Problem: Latency is related to physical distance — data traveling to a distant cloud introduces delay.

Solution: Deploy compute and storage resources **at or near the base station** (at the edge).

- Round-trip time to edge: ~1 ms
- Round-trip time to cloud: ~50–100 ms
- Critical for URLLC (drones, autonomous vehicles, industrial control)

7.3 Public vs. Private Networks

Public network: Shared by all users (B2C, B2B, PPDR). Radio frequency shared → congestion possible during crisis events.

Private network (NPN — Non-Public Network): Own dedicated infrastructure.

Deployment type	Antenna	Spectrum	Core	Data
Public network	Telco	Telco	Telco	Off-site
Public + private APN	Telco	Telco	Telco	Private routing
Public + network slicing	Telco	Telco	Telco (virtual)	Mostly off-site
Public + Telco RAN sharing	Telco	Telco	Shared	Off-site
Fully private network	Private	Dedicated (licensed)	On-premise	On-site

Citymesh analogy:

- Public 5G = shared highway (all users compete for same lanes)
- Crisis situation = ring around Brussels at 9am Monday (congested)
- Private 5G = your own private highway lane — you decide who enters

7.4 5G Slicing Risks (from Citymesh guest lecture)

Risk	Description
Single point of failure	Still uses same physical antenna — if tower fails, all slices fail

Risk	Description
Backhaul dependency	Slicing relies on off-site 5G Core via fiber — cut fiber, slice dies
Shared "pavement"	In extreme congestion, slice isolation can fail — prioritized slice may still experience jitter

7.5 Private 5G vs. 5G Slicing Comparison

Feature	5G Network Slicing	Private 5G Network
Infrastructure	Shared (telco towers, cables)	Dedicated (own mast, radio, server)
Core ("brain")	Off-site cloud	On-site (on-premise)
Disaster resilience	Vulnerable (fiber cut → slice dies)	Standalone — works without uplink
Spectrum	Shared (VIP lane on shared highway)	Exclusive (own private highway)
Control	Limited (telco decides)	Absolute (you decide per device)
Deployment	Static (telco coverage)	Mobile (deploy anywhere, incl. "black zones")

8. Guest Lecture Insights

8.1 Addestino — Real-life 5G Project Cases

Case 1: Private 5G MPN for Warehousing

Context: Telecom operator defining private 5G product strategy for warehousing/logistics.

Warehousing problems: Industry 4.0 automation needs; large outdoor+indoor surfaces; changing radio environment (rack contents); critical IT and OT traffic.

WiFi vs. MPN technical comparison:

Requirement	WiFi capability	MPN (4G/5G) capability
Security	Config-dependent; often neglected	Built-in SIM auth, encryption, slicing
QoS	Best-effort, contention-based	Guaranteed BW, latency, jitter, packet loss
Mobility/handover	Mesh: ~seconds interruption	~ms interruption
Capacity	30–100 devices/AP	10,000 devices/km ²
Coverage	~10m radius per AP	~km radius per base station
Compatibility	Large Wi-Fi device ecosystem	Large 4G ecosystem; 5G still growing

TCO conclusion: MPN makes cost sense for large warehouses (>15,000 m²), especially with outdoor component. Smaller warehouses still favor Wi-Fi on pure cost.

Do warehouses need 5G or is 4G sufficient?

- Low latency (100ms 4G vs. 1ms 5G): **4G OK** for current warehouse automation
- Data rate (1 Gbps 4G vs. 10 Gbps 5G): **4G OK**
- Connection density (10⁴/km² 4G vs. 10⁶ 5G): **4G OK** for current scale
- Reliability (99.99% 4G vs. 99.999% 5G): **4G OK** for most cases
- **Conclusion:** 5G improvements are not yet required for typical warehousing — 4G MPN is sufficient today.

Case 2: 5G for Mission-Critical Industrial Plant

Context: Utility provider with obsolete comms (pagers, DECT, VOIP, TETRA walkie-talkies) — all end-of-support. Must replace within 5 years.

Scenario comparison:

- **Keep existing:** like-for-like replacement of each device type — limited functionality, high ongoing maintenance
- **Private 5G:** single unified platform on one device type; outdoor + indoor deployment; key use cases: digital field ops, push-to-talk, man-down trigger, emergency counting, flexible networks

5G deployment requirements for mission-critical environment:

1. **Safety & regulatory compliance** — certify 5G in safety-regulated plant
2. **Cybersecurity & physical security** — protect against remote compromise, tampering
3. **Electromagnetic compatibility (EMC)** — 5G signals must not interfere with plant instrumentation
4. **Self-sufficiency & operational independence** — must work during external connectivity loss
5. **Future-proofness & organizational acceptance** — avoid vendor lock-in; ensure adoption

Key insight: TCO and use case fit are only part of the decision. In mission-critical environments, regulatory compliance, cybersecurity, EMC, and operational independence become primary decision criteria.

8.2 Citymesh — Private 5G in Practice

Citymesh is a Belgian B2B telecom operator (part of Citymesh Group) offering private 4G/5G networks, IoT, DAS, and TETRA-DMR solutions. Also operates a (mobile) network (DIGI brand).

Network architecture (Citymesh view):

```
Internet ← PDN ← Transit ← CORE ← (Backhaul) ← BBU ← (Fronthaul) ← RRH ← UE
                                     |____ RAN ____|
```

Real-world private 5G use cases:

- **Cosco (Zeebrugge port terminal):** replaced 80+ Wi-Fi APs covering 4G outdoor — bad coverage with containers. Replaced VoIP-on-WiFi with OTT app over private 4G. Operational since 2019.
- **AZ Zeno hospital:** DECT replacement for modern smartphones with scanning, video calling, outside-area calling. Drone transport between hospital campuses. NB-IoT health monitoring bands.
- **Brussels Airport:** Full indoor and outdoor coverage via DAS + leaky feeders. Illustrated by coverage map (purple=deep indoor, red=1st wall, yellow=in-car, green=in-car with booster, blue=outdoor).
- **King's Coronation (London):** Private 5G network deployed along The Mall to support media crews — public network was saturated.
- **Graspop / Dour Festival:** Connected 1,000+ devices over 200,000 m² with max 2 setup points. Temporary deployment by Citymesh Flex division.

9. Current Status (2025)

- Main focus: **eMBB** and **NSA** deployments
 - 5G now available in many populated areas; still rolling out in rural zones
 - Belgium: Proximus 5G coverage expanded significantly (from few spots in 2021 to broad coverage by 2025)
 - Reality check: individual speeds depend heavily on number of concurrent users; peak rates rarely achieved
 - South Korea example: wide 5G coverage but limited new content — hype outpaced reality
 - 5G investment is real and growing, but promise of URLLC and mMTC use cases is still materializing
-

PART 2: SUMMARY TABLES & QUICK REFERENCE

5G Key Parameters

Parameter	Value
Peak DL data rate (eMBB)	10–20 Gbps
Typical user DL rate	100 Mbps
Peak UL data rate	Several Gbps
Latency target (URLLC)	< 1 ms
Reliability (URLLC)	99.999%
Connection density (mMTC)	10 ⁶ /km ²
Max speed supported	500 km/h
Subframe duration	1 ms (fixed)
Radio frame duration	10 ms (fixed)
Subcarrier spacing range	15–240 kHz
Slot duration range	0.0625 – 1 ms

Parameter	Value
FR1 frequency range	450 MHz – 6 GHz
FR2 frequency range	24.25 – 52.6 GHz

Architecture: 5G vs. LTE vs. GSM

Generation	RAN	Core	Key difference
GSM (2G)	BTS + BSC	MSC + HLR/VLR	Circuit-switched voice
LTE (4G)	eNodeB (flat)	EPC (MME+SGW+PGW)	All-IP, flat RAN
5G	gNB / RU+DU+CU	5GC	Service-based, sliced, cloud-native

NR Numerology Quick Reference

μ	SCS (kHz)	Slot (ms)	Slots/subframe	Typical use
0	15	1	1	LTE compatibility, FR1
1	30	0.5	2	FR1 mainstream
2	60	0.25	4	FR1/FR2
3	120	0.125	8	FR2 (mmWave)
4	240	0.0625	16	FR2 reference signals

Network Slicing Enablers

Technology	Role in slicing
SDN	Centralized control — manages which physical resources each slice gets
NFV	Virtualizes network functions — runs per-slice software on shared hardware
MEC	Keeps latency-critical slice processing close to the UE

Private Network Deployment Options

Option	Independence	Cost	QoS guarantee	Security
Public network	None	Low	Best-effort	Low
Private APN	Low	Low-medium	Best-effort	Medium
Network slicing	Medium	Medium	Guaranteed (mostly)	Medium-high
RAN sharing	Medium	Medium	Best-effort	Medium
Fully private network	Full	High	Guaranteed	Highest